Formal component-wise ablation: five backbones, six configurations (tau = 0.4 / 0.5 / 0.6)

Test split: 1,036 queries (544 positive / 492 negative); all values: %. Parentheses: gain versus the same-backbone GMR, pp.

		Operating threshold tau = 0.4									**Operating threshold tau = 0.5**									**Operating threshold tau = 0.6**								
		Null-Set Rejection			*Temporal Localization*				*Overall GMR*		*Null-Set Rejection*			*Temporal Localization*				*Overall GMR*		*Null-Set Rejection*			*Temporal Localization*				*Overall GMR*	
Backbone	**Configuration (cumulative)**	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	1. GMR (coupled baseline)	0.00	70.20	0.00	6.68	**4.64**	9.44	0.88	5.83	2.43	79.27	70.20	67.94	6.68	**4.64**	9.44	0.88	42.50	39.39	**100.00**	70.20	74.94	6.68	**4.64**	9.44	0.88	52.02	49.00
	2. + Global Event Query	**100.00** (+100.00)	**70.33** (+0.13)	**75.00** (+75.00)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+46.19)	49.00 (+46.57)	**100.00** (+20.73)	**70.33** (+0.13)	**75.00** (+7.06)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+9.52)	49.00 (+9.61)	**100.00** (+0.00)	**70.33** (+0.13)	**75.00** (+0.06)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+0.00)	49.00 (+0.00)
	3. + LoRA	**100.00** (+100.00)	70.32 (+0.12)	**75.00** (+75.00)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (+45.95)	48.92 (+46.49)	**100.00** (+20.73)	70.32 (+0.12)	**75.00** (+7.06)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (+9.28)	48.92 (+9.53)	**100.00** (+0.00)	70.32 (+0.12)	**75.00** (+0.06)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (-0.24)	48.92 (-0.08)
	4. + Hard-RoI (no saliency)	**100.00** (+100.00)	70.10 (-0.10)	**75.00** (+75.00)	7.35 (+0.67)	3.48 (-1.16)	11.13 (+1.69)	0.76 (-0.12)	**52.34** (+46.51)	**49.11** (+46.68)	**100.00** (+20.73)	70.10 (-0.10)	**75.00** (+7.06)	7.35 (+0.67)	3.48 (-1.16)	11.13 (+1.69)	0.76 (-0.12)	**52.34** (+9.84)	**49.11** (+9.72)	**100.00** (+0.00)	70.10 (-0.10)	**75.00** (+0.06)	7.35 (+0.67)	3.48 (-1.16)	11.13 (+1.69)	0.76 (-0.12)	**52.34** (+0.32)	**49.11** (+0.11)
	5. + Gaussian-RoI (no saliency)	**100.00** (+100.00)	70.32 (+0.12)	**75.00** (+75.00)	**7.98** (+1.30)	3.88 (-0.76)	**13.26** (+3.82)	0.79 (-0.09)	52.32 (+46.49)	49.10 (+46.67)	**100.00** (+20.73)	70.32 (+0.12)	**75.00** (+7.06)	**7.98** (+1.30)	3.88 (-0.76)	**13.26** (+3.82)	0.79 (-0.09)	52.32 (+9.82)	49.10 (+9.71)	**100.00** (+0.00)	70.32 (+0.12)	**75.00** (+0.06)	**7.98** (+1.30)	3.88 (-0.76)	**13.26** (+3.82)	0.79 (-0.09)	52.32 (+0.30)	49.10 (+0.10)
	6. Full DeRoI-GMR (+6D saliency)	**100.00** (+100.00)	70.26 (+0.06)	**75.00** (+75.00)	7.59 (+0.91)	3.46 (-1.18)	12.40 (+2.96)	**0.98** (+0.10)	51.92 (+46.09)	48.97 (+46.54)	**100.00** (+20.73)	70.26 (+0.06)	**75.00** (+7.06)	7.59 (+0.91)	3.46 (-1.18)	12.40 (+2.96)	**0.98** (+0.10)	51.92 (+9.42)	48.97 (+9.58)	**100.00** (+0.00)	70.26 (+0.06)	**75.00** (+0.06)	7.59 (+0.91)	3.46 (-1.18)	12.40 (+2.96)	**0.98** (+0.10)	51.92 (-0.10)	48.97 (-0.03)
QD-DETR	1. GMR (coupled baseline)	56.71	**77.01**	61.73	14.62	9.47	20.17	7.43	36.51	31.47	79.47	**77.01**	71.48	14.62	9.47	20.17	7.43	46.49	41.47	90.65	**77.01**	73.29	14.62	9.47	20.17	7.43	50.68	46.04
	2. + Global Event Query	80.28 (+23.57)	76.44 (-0.57)	71.11 (+9.38)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	46.60 (+10.09)	41.78 (+10.31)	92.48 (+13.01)	76.44 (-0.57)	74.29 (+2.81)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	51.54 (+5.05)	46.91 (+5.44)	96.54 (+5.89)	76.44 (-0.57)	75.04 (+1.75)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	53.09 (+2.41)	48.44 (+2.40)
	3. + LoRA	80.49 (+23.78)	76.48 (-0.53)	71.22 (+9.49)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	46.00 (+9.49)	41.58 (+10.11)	92.48 (+13.01)	76.48 (-0.53)	74.41 (+2.93)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	51.00 (+4.51)	46.73 (+5.26)	96.54 (+5.89)	76.48 (-0.53)	75.16 (+1.87)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	52.54 (+1.86)	48.26 (+2.22)
	4. + Hard-RoI (no saliency)	**84.35** (+27.64)	76.21 (-0.80)	72.43 (+10.70)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	11.11 (+3.68)	**48.38** (+11.87)	**43.64** (+12.17)	**93.90** (+14.43)	76.21 (-0.80)	**74.58** (+3.10)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	11.11 (+3.68)	**52.16** (+5.67)	**47.48** (+6.01)	**97.56** (+6.91)	76.21 (-0.80)	**75.47** (+2.18)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	11.11 (+3.68)	**53.70** (+3.02)	**48.96** (+2.92)
	5. + Gaussian-RoI (no saliency)	**84.35** (+27.64)	76.18 (-0.83)	**72.49** (+10.76)	15.83 (+1.21)	9.33 (-0.14)	23.87 (+3.70)	11.82 (+4.39)	48.27 (+11.76)	43.50 (+12.03)	93.70 (+14.23)	76.18 (-0.83)	74.47 (+2.99)	15.83 (+1.21)	9.33 (-0.14)	23.87 (+3.70)	11.82 (+4.39)	52.03 (+5.54)	47.38 (+5.91)	**97.56** (+6.91)	76.18 (-0.83)	75.41 (+2.12)	15.83 (+1.21)	9.33 (-0.14)	23.87 (+3.70)	11.82 (+4.39)	53.57 (+2.89)	48.87 (+2.83)
	6. Full DeRoI-GMR (+6D saliency)	81.50 (+24.79)	76.21 (-0.80)	71.54 (+9.81)	16.18 (+1.56)	9.75 (+0.28)	23.77 (+3.60)	**12.26** (+4.83)	47.35 (+10.84)	42.37 (+10.90)	92.68 (+13.21)	76.21 (-0.80)	74.21 (+2.73)	16.18 (+1.56)	9.75 (+0.28)	23.77 (+3.60)	**12.26** (+4.83)	51.77 (+5.28)	47.02 (+5.55)	96.95 (+6.30)	76.21 (-0.80)	75.18 (+1.89)	16.18 (+1.56)	9.75 (+0.28)	23.77 (+3.60)	**12.26** (+4.83)	53.50 (+2.82)	48.67 (+2.63)
CG-DETR	1. GMR (coupled baseline)	23.17	76.40	35.46	18.66	12.65	24.90	11.98	22.84	17.07	50.41	76.40	57.47	18.66	12.65	24.90	11.98	34.35	29.14	78.46	76.40	71.09	18.66	12.65	24.90	11.98	46.66	41.53
	2. + Global Event Query	**89.23** (+66.06)	76.70 (+0.30)	**73.78** (+38.32)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	**50.55** (+27.71)	**45.67** (+28.60)	**96.95** (+46.54)	76.70 (+0.30)	75.36 (+17.89)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	53.49 (+19.14)	48.76 (+19.62)	98.98 (+20.52)	76.70 (+0.30)	75.68 (+4.59)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	54.34 (+7.68)	49.62 (+8.09)
	3. + LoRA	87.80 (+64.63)	**76.73** (+0.33)	72.97 (+37.51)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	49.69 (+26.85)	44.96 (+27.89)	**96.95** (+46.54)	**76.73** (+0.33)	**75.42** (+17.95)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	53.41 (+19.06)	48.74 (+19.60)	98.98 (+20.52)	**76.73** (+0.33)	75.68 (+4.59)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	54.25 (+7.59)	49.59 (+8.06)
	4. + Hard-RoI (no saliency)	87.80 (+64.63)	76.64 (+0.24)	73.03 (+37.57)	18.93 (+0.27)	13.19 (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	50.02 (+27.18)	45.00 (+27.93)	**96.95** (+46.54)	76.64 (+0.24)	75.36 (+17.89)	18.93 (+0.27)	13.19 (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	**53.67** (+19.32)	**48.80** (+19.66)	**99.19** (+20.73)	76.64 (+0.24)	75.66 (+4.57)	18.93 (+0.27)	13.19 (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	54.54 (+7.88)	**49.73** (+8.20)
	5. + Gaussian-RoI (no saliency)	84.35 (+61.18)	76.57 (+0.17)	72.11 (+36.65)	**20.36** (+1.70)	**13.28** (+0.63)	**28.10** (+3.20)	12.36 (+0.38)	48.84 (+26.00)	43.62 (+26.55)	95.73 (+45.32)	76.57 (+0.17)	75.24 (+17.77)	**20.36** (+1.70)	**13.28** (+0.63)	**28.10** (+3.20)	12.36 (+0.38)	53.37 (+19.02)	48.43 (+19.29)	98.98 (+20.52)	76.57 (+0.17)	**75.80** (+4.71)	**20.36** (+1.70)	**13.28** (+0.63)	**28.10** (+3.20)	12.36 (+0.38)	**54.60** (+7.94)	**49.73** (+8.20)
	6. Full DeRoI-GMR (+6D saliency)	89.02 (+65.85)	76.72 (+0.32)	73.55 (+38.09)	19.33 (+0.67)	13.08 (+0.43)	25.77 (+0.87)	11.87 (-0.11)	50.52 (+27.68)	45.59 (+28.52)	**96.95** (+46.54)	76.72 (+0.32)	75.30 (+17.83)	19.33 (+0.67)	13.08 (+0.43)	25.77 (+0.87)	11.87 (-0.11)	53.60 (+19.25)	48.79 (+19.65)	**99.19** (+20.73)	76.72 (+0.32)	75.72 (+4.63)	19.33 (+0.67)	13.08 (+0.43)	25.77 (+0.87)	11.87 (-0.11)	54.49 (+7.83)	49.71 (+8.18)
EaTR	1. GMR (coupled baseline)	48.58	**76.56**	57.11	16.22	10.64	21.42	9.59	33.32	28.34	66.67	**76.56**	65.60	16.22	10.64	21.42	9.59	40.85	36.00	87.20	**76.56**	73.46	16.22	10.64	21.42	9.59	49.56	44.86
	2. + Global Event Query	82.52 (+33.94)	75.73 (-0.83)	71.35 (+14.24)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	47.14 (+13.82)	42.52 (+14.18)	95.53 (+28.86)	75.73 (-0.83)	75.26 (+9.66)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	52.83 (+11.98)	48.29 (+12.29)	98.98 (+11.78)	75.73 (-0.83)	**75.68** (+2.22)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	54.22 (+4.66)	49.56 (+4.70)
	3. + LoRA	83.94 (+35.36)	75.76 (-0.80)	71.76 (+14.65)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	47.78 (+14.46)	43.18 (+14.84)	95.53 (+28.86)	75.76 (-0.80)	74.96 (+9.36)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	52.81 (+11.96)	48.22 (+12.22)	98.98 (+11.78)	75.76 (-0.80)	75.56 (+2.10)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	54.24 (+4.68)	49.57 (+4.71)
	4. + Hard-RoI (no saliency)	**85.98** (+37.40)	76.02 (-0.54)	72.12 (+15.01)	**17.16** (+0.94)	11.43 (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	**48.71** (+15.39)	**44.08** (+15.74)	95.93 (+29.26)	76.02 (-0.54)	74.98 (+9.38)	**17.16** (+0.94)	11.43 (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	52.95 (+12.10)	48.34 (+12.34)	**99.19** (+11.99)	76.02 (-0.54)	75.60 (+2.14)	**17.16** (+0.94)	11.43 (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	54.25 (+4.69)	49.65 (+4.79)
	5. + Gaussian-RoI (no saliency)	**85.98** (+37.40)	75.99 (-0.57)	**72.37** (+15.26)	16.25 (+0.03)	10.51 (-0.13)	22.19 (+0.77)	**10.21** (+0.62)	48.49 (+15.17)	44.04 (+15.70)	**96.54** (+29.87)	75.99 (-0.57)	**75.40** (+9.80)	16.25 (+0.03)	10.51 (-0.13)	22.19 (+0.77)	**10.21** (+0.62)	52.92 (+12.07)	**48.56** (+12.56)	**99.19** (+11.99)	75.99 (-0.57)	75.60 (+2.14)	16.25 (+0.03)	10.51 (-0.13)	22.19 (+0.77)	**10.21** (+0.62)	53.86 (+4.30)	49.52 (+4.66)
	6. Full DeRoI-GMR (+6D saliency)	85.37 (+36.79)	76.03 (-0.53)	72.29 (+15.18)	17.01 (+0.79)	**11.44** (+0.80)	22.48 (+1.06)	9.63 (+0.04)	48.58 (+15.26)	43.90 (+15.56)	96.14 (+29.47)	76.03 (-0.53)	75.32 (+9.72)	17.01 (+0.79)	**11.44** (+0.80)	22.48 (+1.06)	9.63 (+0.04)	**53.12** (+12.27)	48.53 (+12.53)	**99.19** (+11.99)	76.03 (-0.53)	75.66 (+2.20)	17.01 (+0.79)	**11.44** (+0.80)	22.48 (+1.06)	9.63 (+0.04)	**54.26** (+4.70)	**49.69** (+4.83)
FlashVTG	1. GMR (coupled baseline)	32.72	74.30	44.41	25.43	15.17	34.94	17.86	28.10	21.75	70.93	74.30	66.35	25.43	15.17	34.94	17.86	43.57	38.27	85.16	74.30	71.56	25.43	15.17	34.94	17.86	49.41	44.30
	2. + Global Event Query	88.01 (+55.29)	75.26 (+0.96)	73.08 (+28.67)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	50.48 (+22.38)	45.43 (+23.68)	94.72 (+23.79)	75.26 (+0.96)	74.38 (+8.03)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	53.01 (+9.44)	48.11 (+9.84)	96.95 (+11.79)	75.26 (+0.96)	74.59 (+3.03)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	53.71 (+4.30)	48.90 (+4.60)
	3. + LoRA	88.82 (+56.10)	**75.32** (+1.02)	73.45 (+29.04)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	50.80 (+22.70)	45.74 (+23.99)	94.31 (+23.38)	**75.32** (+1.02)	74.18 (+7.83)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	52.78 (+9.21)	47.90 (+9.63)	96.75 (+11.59)	**75.32** (+1.02)	**74.67** (+3.11)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	53.68 (+4.27)	48.83 (+4.53)
	4. + Hard-RoI (no saliency)	90.04 (+57.32)	75.25 (+0.95)	73.34 (+28.93)	26.24 (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	18.27 (+0.41)	51.67 (+23.57)	46.41 (+24.66)	**95.73** (+24.80)	75.25 (+0.95)	74.64 (+8.29)	26.24 (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	18.27 (+0.41)	53.68 (+10.11)	48.62 (+10.35)	**97.36** (+12.20)	75.25 (+0.95)	74.61 (+3.05)	26.24 (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	18.27 (+0.41)	54.19 (+4.78)	49.20 (+4.90)
	5. + Gaussian-RoI (no saliency)	**90.45** (+57.73)	75.31 (+1.01)	**73.55** (+29.14)	**26.29** (+0.86)	16.10 (+0.93)	36.21 (+1.27)	**19.05** (+1.19)	**52.01** (+23.91)	**46.65** (+24.90)	**95.73** (+24.80)	75.31 (+1.01)	**74.70** (+8.35)	**26.29** (+0.86)	16.10 (+0.93)	36.21 (+1.27)	**19.05** (+1.19)	**53.88** (+10.31)	**48.73** (+10.46)	**97.36** (+12.20)	75.31 (+1.01)	74.61 (+3.05)	**26.29** (+0.86)	16.10 (+0.93)	36.21 (+1.27)	**19.05** (+1.19)	**54.34** (+4.93)	**49.25** (+4.95)
	6. Full DeRoI-GMR (+6D saliency)	90.24 (+57.52)	75.29 (+0.99)	73.51 (+29.10)	26.13 (+0.70)	15.83 (+0.66)	35.68 (+0.74)	18.46 (+0.60)	51.84 (+23.74)	46.54 (+24.79)	**95.73** (+24.80)	75.29 (+0.99)	74.64 (+8.29)	26.13 (+0.70)	15.83 (+0.66)	35.68 (+0.74)	18.46 (+0.60)	53.72 (+10.15)	48.63 (+10.36)	**97.36** (+12.20)	75.29 (+0.99)	74.61 (+3.05)	26.13 (+0.70)	15.83 (+0.66)	35.68 (+0.74)	18.46 (+0.60)	54.22 (+4.81)	49.21 (+4.91)

Formal rows only: configurations 1-6. The five fusion-enabled diagnostic runs (execution row 7) are published separately.

Added rows: value (delta versus configuration-1 GMR, same backbone / split / threshold). Gains subtract displayed two-decimal values.

Bold: within-backbone best; underline: second distinct best (ties retained). Fixed epoch 4; baseline frozen; seed 2026; no test selection.

Native FlashVTG soft score calibration / rounding / NMS is retained. Matched CG/EaTR GMR exports differ from historical mAP by 0.01/0.02 pp.